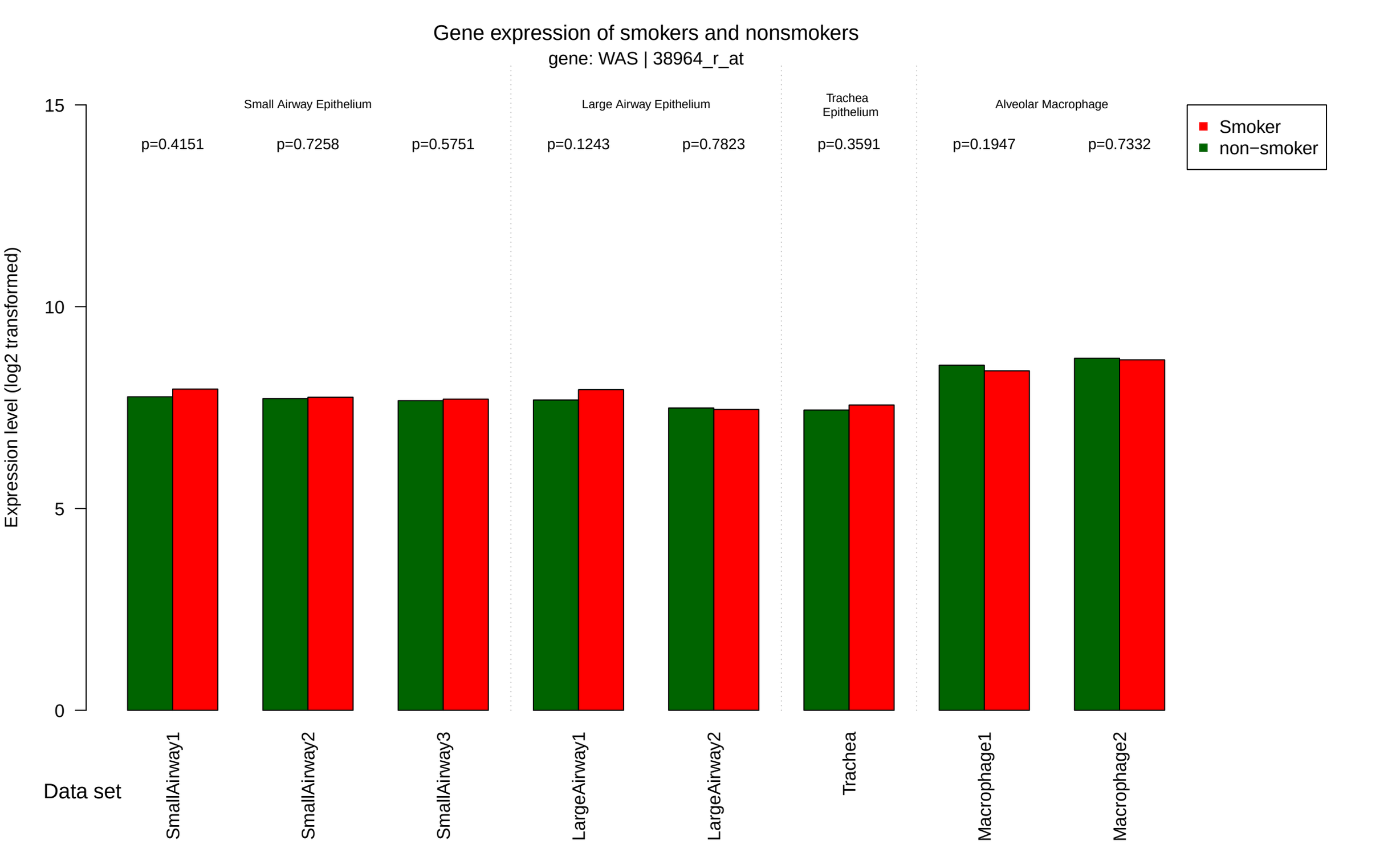
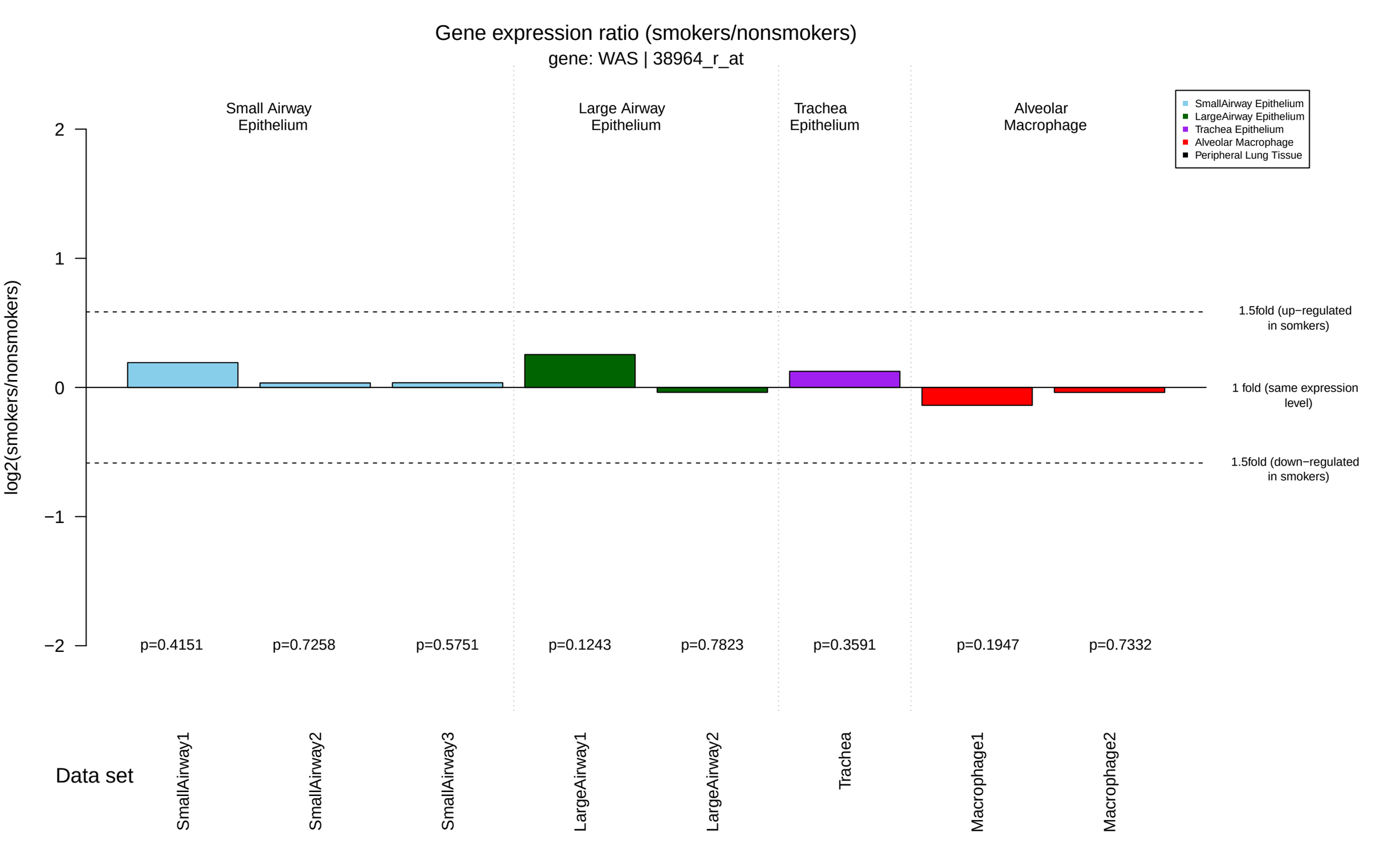
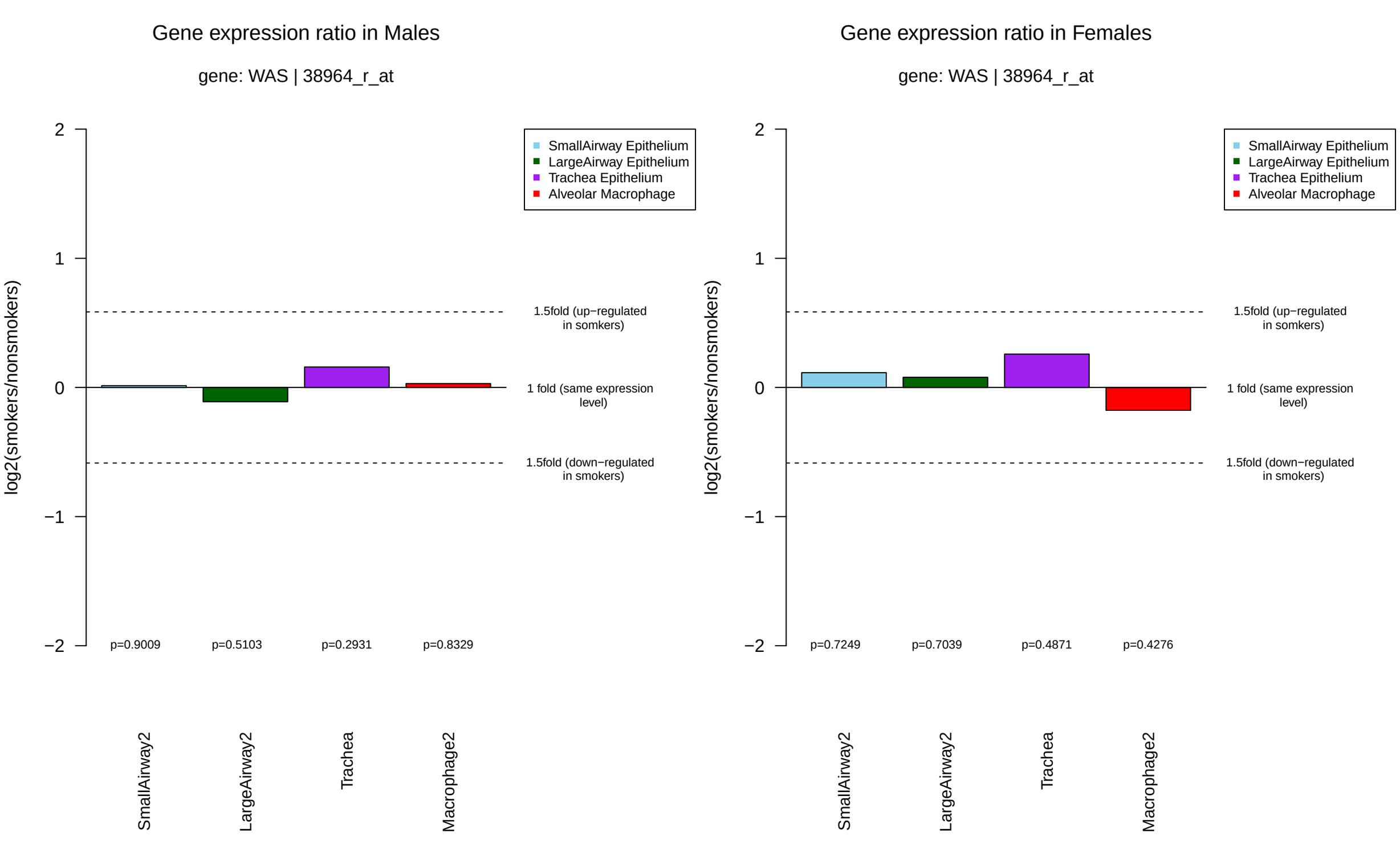
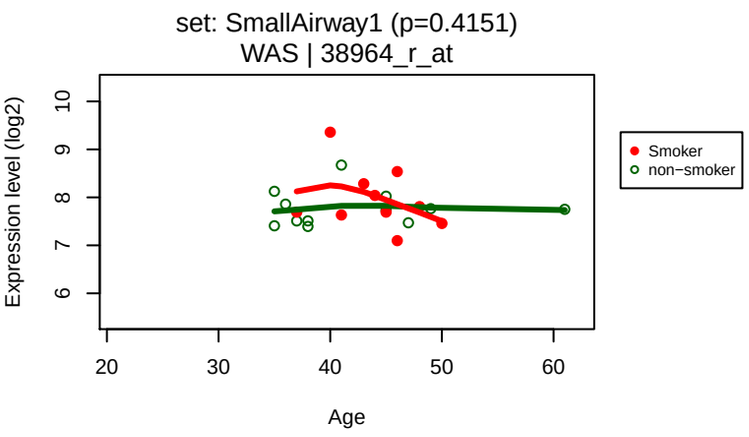
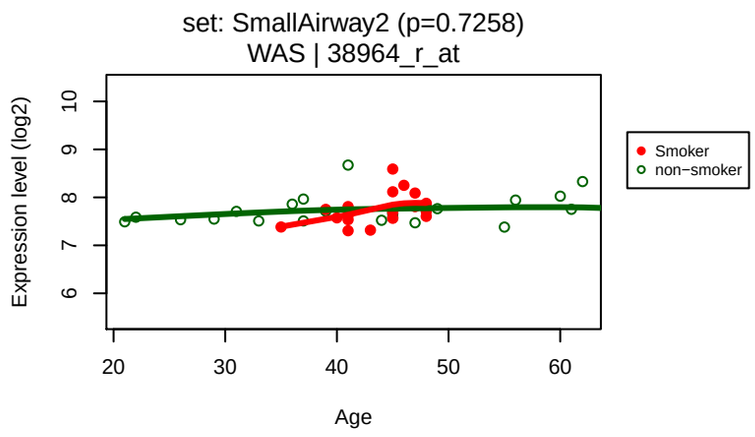
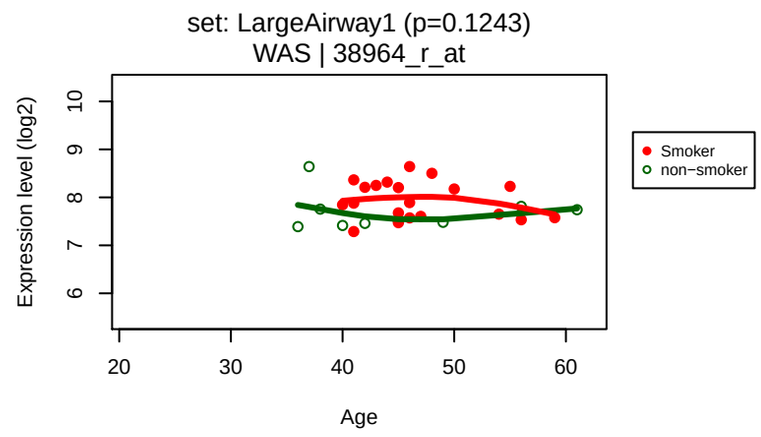

Gene expression of smokers and nonsmokers
gene: WAS | 38964_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4151
p=0.7258
p=0.5751
p=0.1243
p=0.7823
p=0.3591
p=0.1947
p=0.7332
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

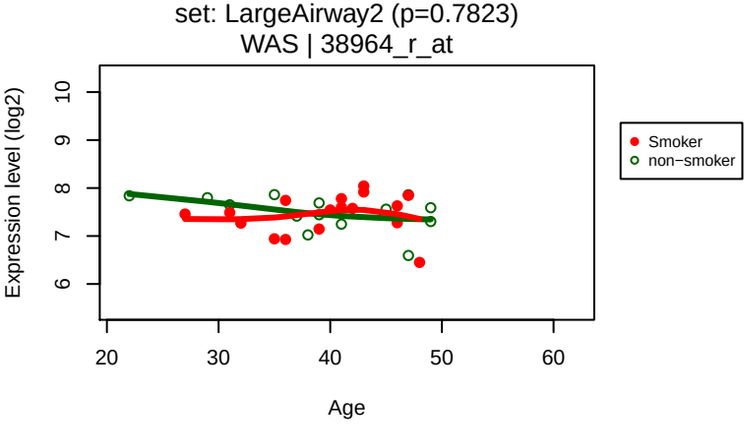

Gene expression ratio (smokers/nonsmokers)
gene: WAS | 38964_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4151
p=0.7258
p=0.5751
p=0.1243
p=0.7823
p=0.3591
p=0.1947
p=0.7332
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

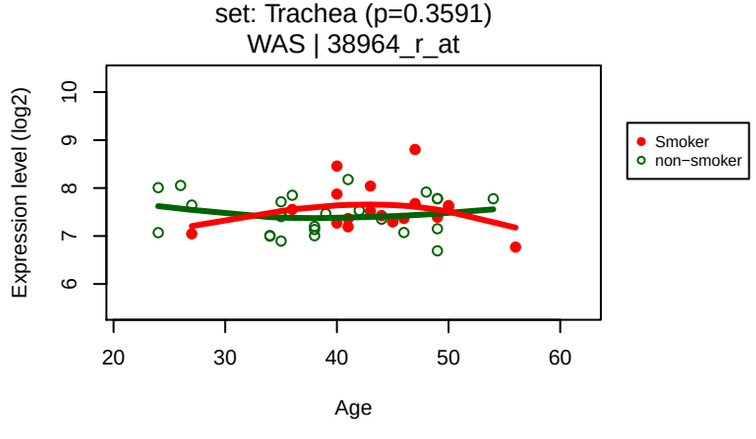

Gene expression ratio in Males
gene: WAS | 38964_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9009
p=0.5103
p=0.2931
p=0.8329
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAS | 38964_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7249
p=0.7039
p=0.4871
p=0.4276
SmallAirway2
LargeAirway2
Trachea
Macrophage2

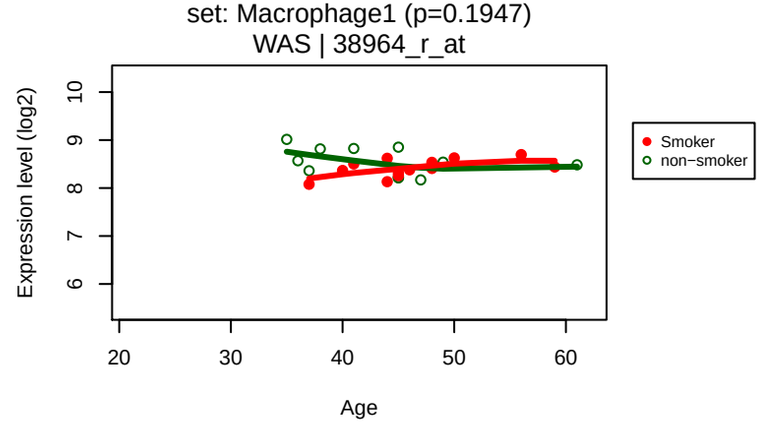

set: SmallAirway1 (p=0.4151)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker

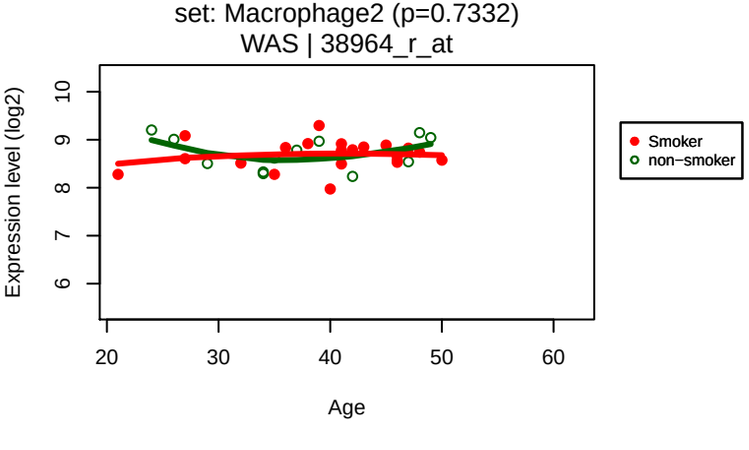

set: SmallAirway2 (p=0.7258)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker

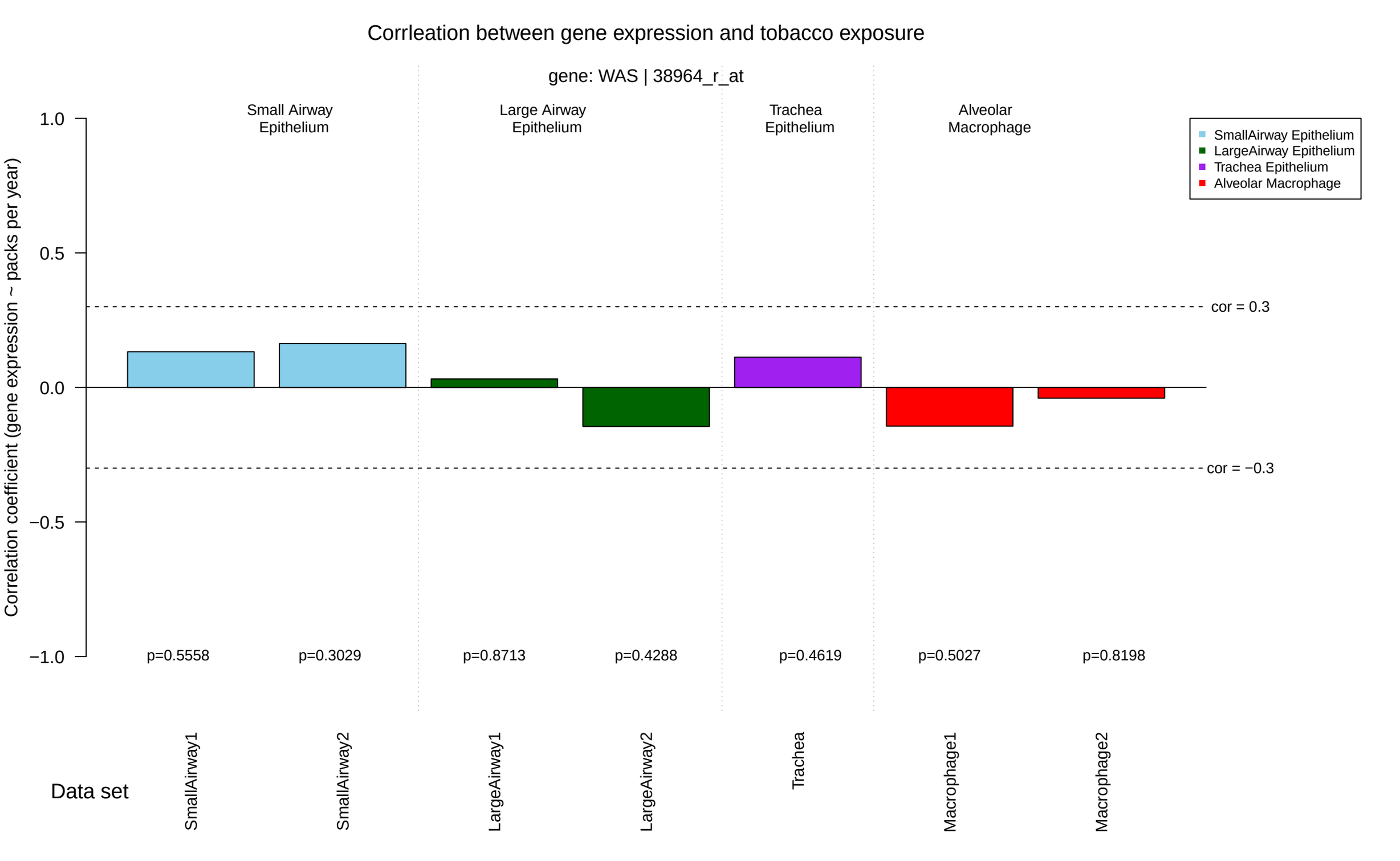

set: LargeAirway1 (p=0.1243)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7823)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3591)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1947)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7332)
WAS | 38964_r_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAS | 38964_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5558
p=0.3029
p=0.8713
p=0.4288
p=0.4619
p=0.5027
p=0.8198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2